

Adem Yahmadi

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PROFILE

Computer Science student at TUM building backend systems, optimization tools, and applied AI software. Interested in turning mathematical and engineering problems into reliable software. Seeking a working student position.

EDUCATION

• Technical University of Munich

Bachelor of Science in Informatics

2023 – Present
Munich, Germany

• Lycée Hrairia 2

Scientific Baccalaureate, Mathematics

2019 – 2023
Tunis, Tunisia

EXPERIENCE

• TUfast Eco Team

Driving Strategy Engineer

2024 – 2025
Technical University of Munich

- Designed the race strategy optimizer for the lux025 solar car (Bridgestone World Solar Challenge 2025), planning speed profiles over 3,000 km under energy and race constraints.
- Extended the PyTorch vehicle model to optimize driving speed with gradient descent, and integrated multiple optimization solvers (IPOPT, ParOpt, ALPSO).
- Built a PySide6 GUI for running optimizations and wrote unit tests for the control-stop simulation (pytest, GitLab CI).

PROJECTS

• InfraLens – Proxmox Cluster Visualizer & AI Assistant

Python, FastAPI, React, Tauri, Proxmox API, Ollama, SSH

[Live](#) [Code](#)

- Built a Tauri/React desktop app that connects to Proxmox clusters, visualizes nodes, VMs, and containers as an interactive topology graph, and monitors resources in real time.
- Implemented a FastAPI backend for VM and container controls and SSH-based service discovery.
- Integrated a local Ollama LLM that uses live cluster state for natural-language infrastructure diagnostics.

• Vertex – Intelligent Document Scanner

Python, FastAPI, OpenCV, React, Docker

[Live](#) [Code](#)

- Designed a scanning pipeline combining deep learning (DocAligner) and OpenCV for edge detection, perspective correction, and shadow removal.
- Shipped a CLI and React web app that convert photos into full-resolution PDFs; deployed with Docker on Render.

• mdtopdf – Markdown to PDF Converter

Node.js, Express, KaTeX, Mermaid, Supabase, Docker

[Live](#) [Code](#)

- Created a CLI and web app that convert Markdown to print-ready PDFs, with LaTeX math (KaTeX), Mermaid diagrams, and themed syntax highlighting for 200+ languages.
- Built a live-preview editor with PDF export, Supabase auth, and cloud storage; deployed with Docker on Render.

UNIVERSITY PROJECTS

• Motion Planning for Autonomous Vehicles (CommonRoad)

Python, CommonRoad, Shapely, Matplotlib

WS 2025/26

[Paper](#)

- Converted drone-recorded traffic data into CommonRoad simulation scenarios for benchmarking autonomous vehicle motion planners.
- Developed a validation tool with spatial indexing to detect collisions, off-road driving, and physically infeasible motion, with visualizations for debugging.

• Cache Simulator – Computer Architecture Lab

C, C++, SystemC

SS 2024

[Code](#)

- Built a cycle-accurate cache simulator in a team of three, with a C trace parser and a SystemC/C++ hardware model.
- Implemented direct-mapped and fully associative (LRU) caches, comparing hit rate against estimated gate count.

SKILLS

- Programming:** Python, Java, C, C++, JavaScript
- Frameworks & Libraries:** PyTorch, FastAPI, React, Node.js, NumPy, OpenCV
- Tools & Infrastructure:** Linux, Git, Docker, GitLab CI, pytest, SSH, Ollama
- Languages:** German (native), English (C1), Arabic (native), French (C1)